South Dakota Manufacturing Supply Chain Mapping and Dashboard Project

Portfolio-ready report summary | Coyote Business Consulting / SDMTS

Project Overview

South Dakota Manufacturing and Technology Solutions (SDMTS) engaged a Coyote Business Consulting team to support its Supply Chain Optimization and Intelligence Network (SCOIN) work. The project focused on creating a usable map and dashboard of South Dakota manufacturers, improving the usefulness of firm-level source data, and translating mapped patterns into practical supply-chain and economic-development insights.

Role Note

This was a team consulting project with shared technical and client-facing responsibilities. My primary contributions centered on interpreting the analysis, developing the written and presentation deliverables, and translating the findings into client-facing recommendations. The project is included here as an example of applied data consulting, supply-chain mapping, and regional economic analysis.

Client Problem

The client needed a way to view, filter, and interpret manufacturing firm records across South Dakota. The work required moving beyond a static company list toward map-ready records, usable filters, confidence scoring, and client-facing tools that could support conversations with manufacturers and public-sector stakeholders.

Technical Workflow

- Data cleanup and standardization: the team reviewed D&B; Hoovers records, removed duplicate records, normalized fields, and prepared the data for mapping and dashboard use.
- Existence and validity scoring: records were assessed using signals such as phone number, URL, D-U-N-S number, address, and evidence of active-business status. These signals helped produce a confidence-oriented view of the data rather than treating every record as equally reliable.
- Geocoding and map readiness: company addresses were converted into latitude/longitude coordinates, then checked for out-of-state or implausible coordinates that needed manual review.
- Industry classification: three-digit NAICS groupings were used to reduce complexity while still preserving meaningful industry-level differences in the map and dashboard.
- Mapping deliverables: the team used Excel Power Map and BatchGeo to create complementary ways to view the geographic distribution of manufacturers.
- Dashboard development: an Excel dashboard was structured around slicers, search functionality, pivot-driven charts, and a verify-business workflow so users could retrieve and manipulate records in a familiar tool environment.

Why Excel and Mapping Tools Were Used

The project evaluated tool accessibility from the standpoint of the likely end users. Excel was a practical choice because it was familiar, widely available, and capable of supporting dashboard logic through tables, pivot tables, slicers, and macro-enabled workflows. BatchGeo provided a browser-accessible mapping option, while Excel

Power Map provided a more integrated map experience for users working directly with the workbook.

Technical Element	Portfolio Signal
Data cleanup	Ability to move from raw firm-level records to structured analysis inputs
Validation logic	Understanding of data quality, false positives, and confidence scoring
Geocoding	Preparation of map-ready data and manual review of location errors
NAICS grouping	Industry classification and aggregation logic
Dashboard design	Client-facing analytics and usable decision-support tools
Gap analysis	Translation of mapped data into strategic economic-development questions

Selected Findings and Interpretation

The mapping workflow supported a gap-analysis style interpretation. Cities could be grouped by concentration of manufacturers, and industry-specific maps could highlight where relevant suppliers were clustered or absent. This made the tool useful not only for locating firms, but also for asking follow-up questions about co-location, logistics, infrastructure, and targeted industrial development.

Limitations

- The original data source was a third-party business database, so the team could not guarantee completeness or real-time accuracy.
- Public web signals can help flag likely record quality problems, but they are not perfect substitutes for official confirmation or direct outreach.
- Macro-enabled Excel tools are useful for desktop users but are harder to embed directly in a public website.
- The portfolio workbook included publicly is a redacted demonstration and does not disclose raw firm-level working data.

Skills Demonstrated

- Applied data consulting
- Supply-chain mapping
- Excel dashboard logic
- Data cleanup and validation thinking
- Geographic and industry classification
- Client-facing writing and presentation
- Regional economic-development interpretation